

Size Matters

Who are we?

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Creator of DaCy - Best Performing
Danish Language Model



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Language Model



Language models

Language models like the model **BERT** **ORG** trained by **Devlin** **PERSON** can be trained to recognize entities.

- It is a statistical model, which is trained to learn patterns in text data
- It generalizes across tasks
- Speech Recognition
 - “I ate a cherry” vs. “Eye eight uh Jerry”
- Answering questions from text (question answering)
- Writing poems and acting as your everyday chatbots (text generation)
- Etc.

GPT-3 model

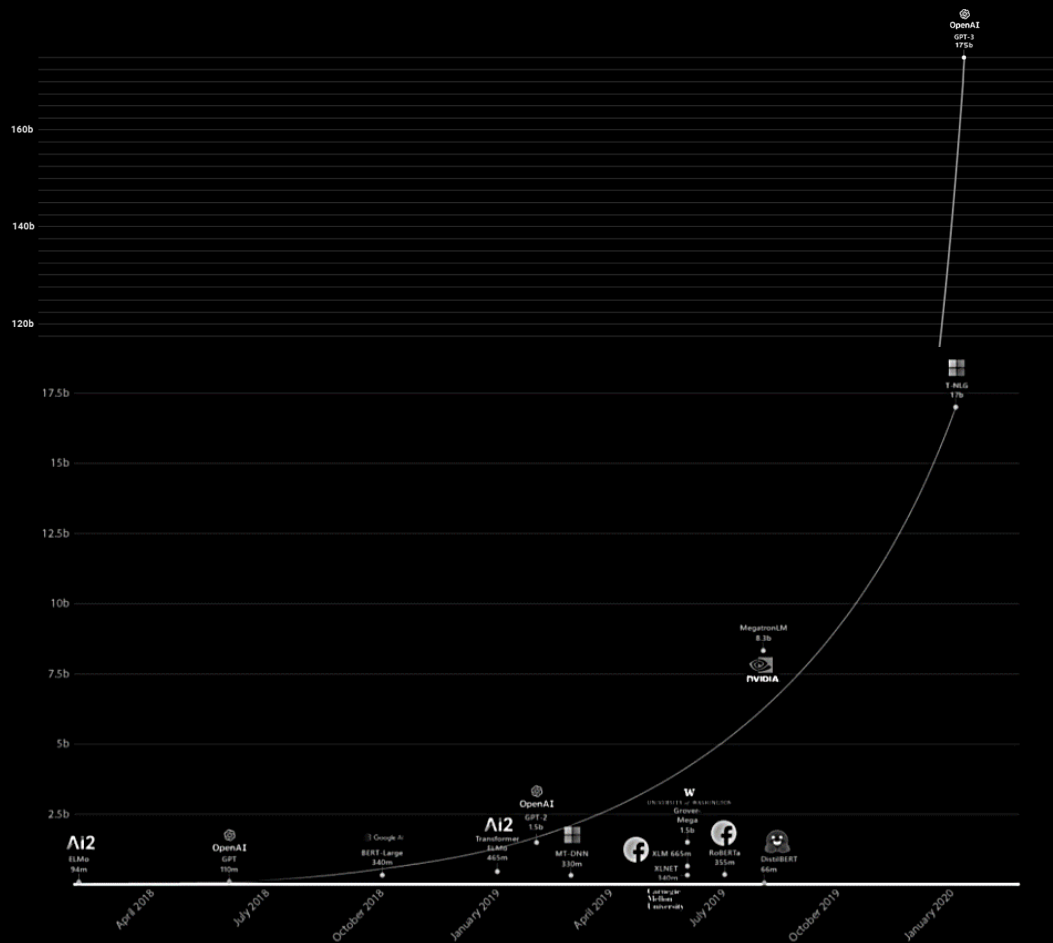
Generative
Created by
Pretrained

- Common Crawl
- Wikipedia
- Etc.

The Danish
words.

State-of-the-art

- Text generation
- Question answering
- Text summarization



How does it work?

GPT-3 - Word predictions

I _____

Bread - Wikipedia

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en.wikipedia.org/wiki/Bread



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The Free Encyclopedia

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Bread



From Wikipedia, the free encyclopedia

For other uses, see [Bread \(disambiguation\)](#).

Bread is a **staple food** prepared from a **dough** of **flour** and **water**, usually by **baking**. Throughout **recorded history** it has been a

Bread



Bread - Wikipedia

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en.wikipedia.org/wiki/Bread

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why is bread so fluffy?

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The rapid expansion of steam produced during baking leavens the bread, which is as simple as it is unpredictable. Steam-leavening is unpredictable since the steam is not produced until the bread is baked. Steam leavening happens regardless of the raising agents (baking soda, yeast, baking powder, sour dough, beaten egg white) included in the mix. The leavening agent either contains air bubbles or generates carbon dioxide. The heat vaporises the water from the inner surface of the bubbles within the dough. The steam expands and makes the bread rise. This is the main factor in the rising of bread once it has been put in the oven.[55] CO2 generation, on its own, is too small to account for the rise. Heat kills bacteria or yeast at an early stage, so the CO2 generation is stopped.

(score: 239.48)

[go to location](#)



GPT Playground



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Default (Ari... ▾

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fx

	A	B	C	D
1	State Name	Population		
2	Illinois	12.67M		
3	California	39.51M		
4	Ohio	11.69M		
5	Michigan			
6				



GPT Playground



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10 ▾

fx

	A	B	C	D
1	State Name	Population	Founded Date	
2	Illinois	12.67M	1818	
3	California	39.51M	1850	
4	Ohio	11.69M	1803	
5	Alaska	603K	1906	
6				

What would you like to know?

How many users have signed up since the start of 2020?

```
1 SELECT
2   count(id)
3 FROM
4   users
5 WHERE
6   created_at > '2020-01-01'
```


Instruction:

Given an input question, respond with syntactically correct PostgreSQL.
Be creative but the SQL must be correct.

Ask GPT-3:

Get GPT-3 Response

Instruction:

Given an input question, respond with syntactically correct PostgreSQL.
Be creative but the SQL must be correct.

Ask GPT-3:

how many users signed up in the

Get GPT-3 Response

Input: how many users signed up in the past month?

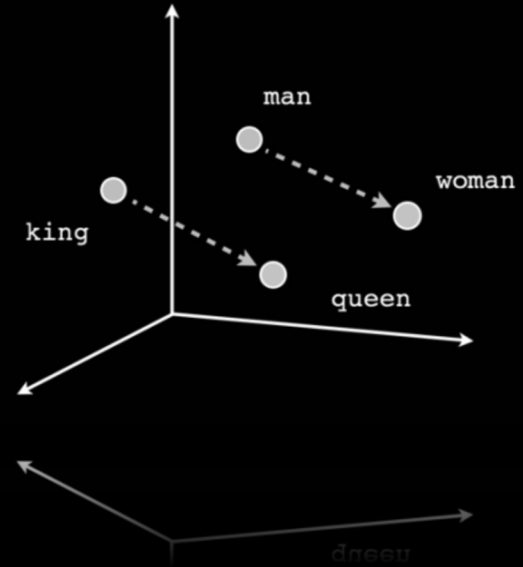
GPT-3 Response: `SELECT COUNT(*) FROM users
WHERE signup_time > now() - interval '1 month'`

Run SQL

GPT-3's Representation of Language

Representation of Language

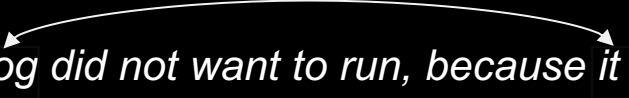
- Numerical word representations
 - Words as vectors, Word Embeddings
 - Word2Vec
 - $\text{King} - \text{Man} + \text{Woman} = \text{Queen}$



Representation of Language cont.

- Word Embeddings
 - Static
 - *“Can companies litter the environment”*
 - Question or statement?
- Contextual Word Embeddings
 - Positional Encodings
 - *Attention Is All You Need* (Vaswani et al. 2017)

“The dog did not want to run, because it was too tired.”



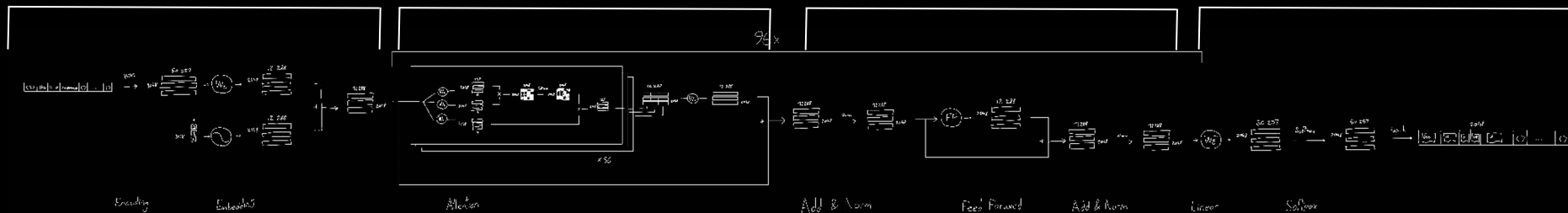
GPT-3 Pipeline

Text
to
Vector

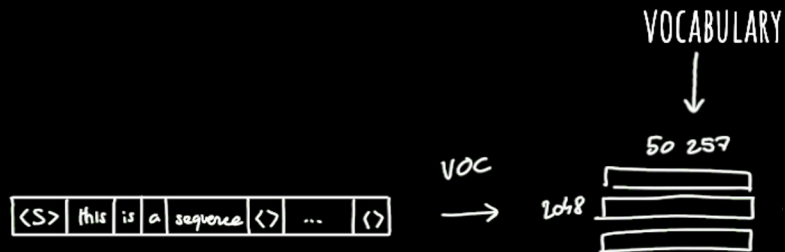
Contextual Update
Using
Attention

Representation
Update

Vector
to
Word Prediction



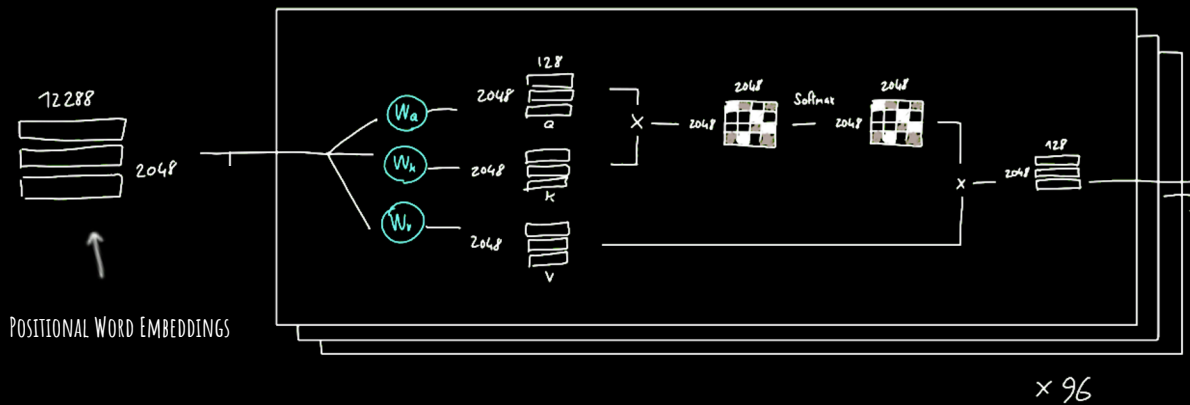
Text to Vector



Encoding

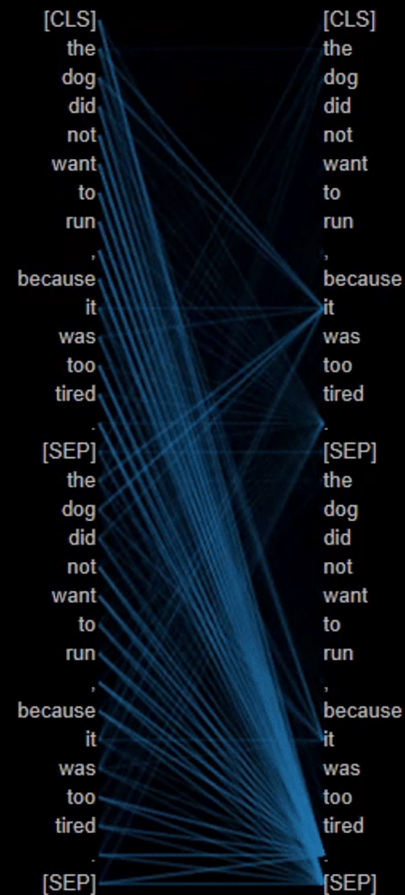
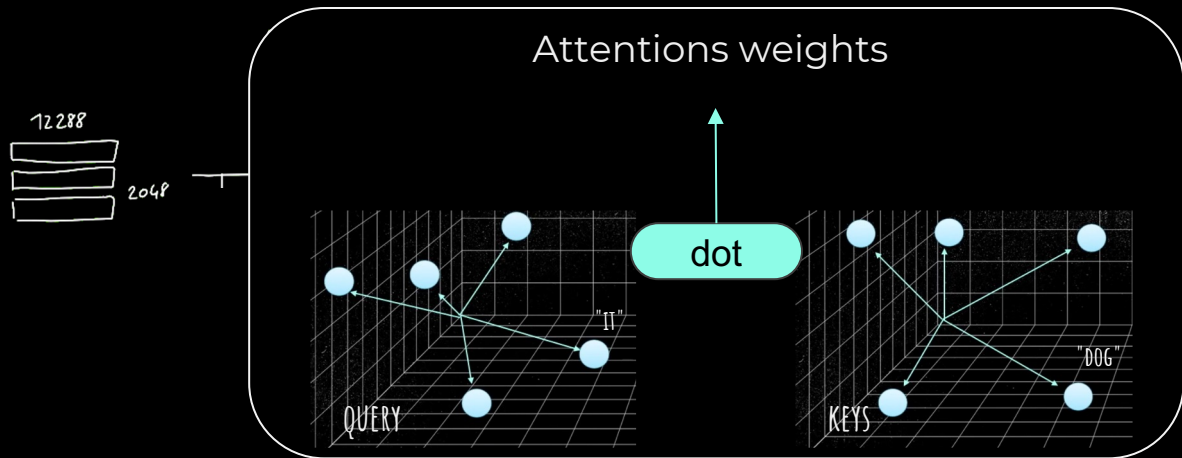
Embedding

Contextual Update Using Attention

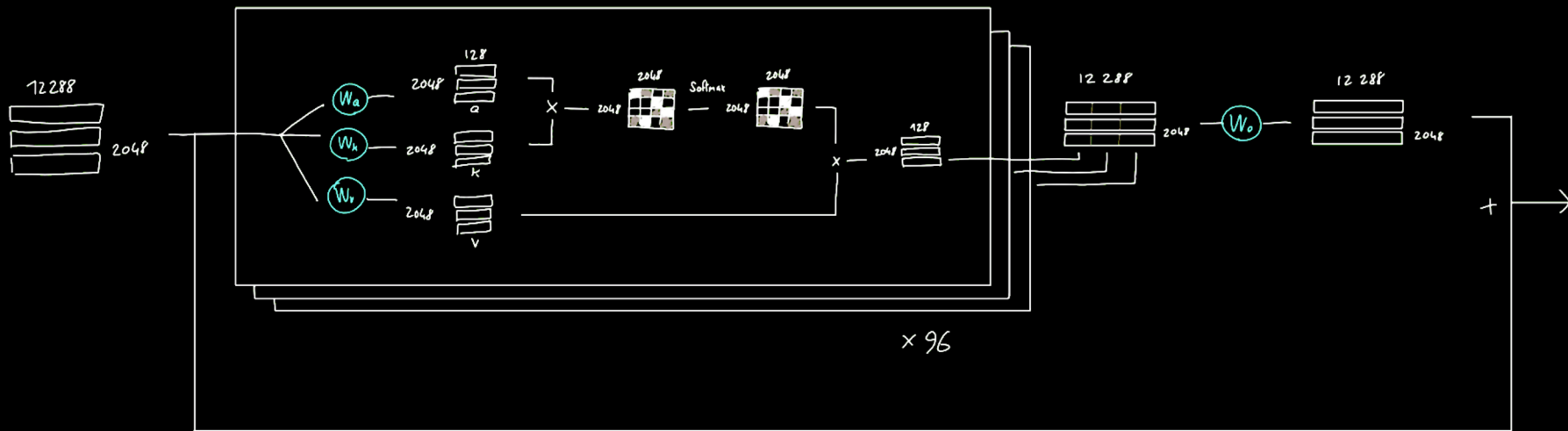


Attention

Contextual Update Using Attention

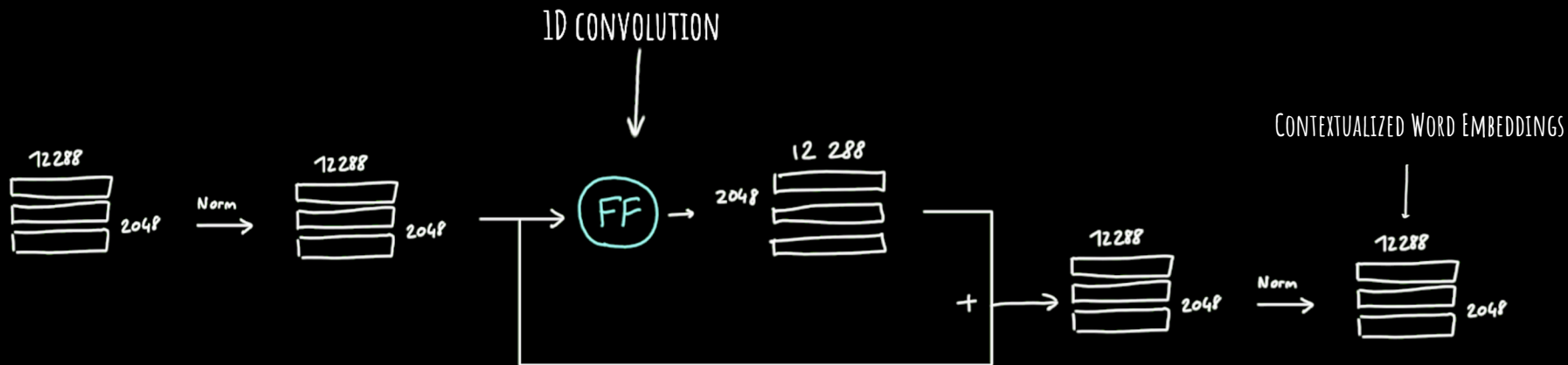


Contextual Update Using Attention



Attention

Representation Update

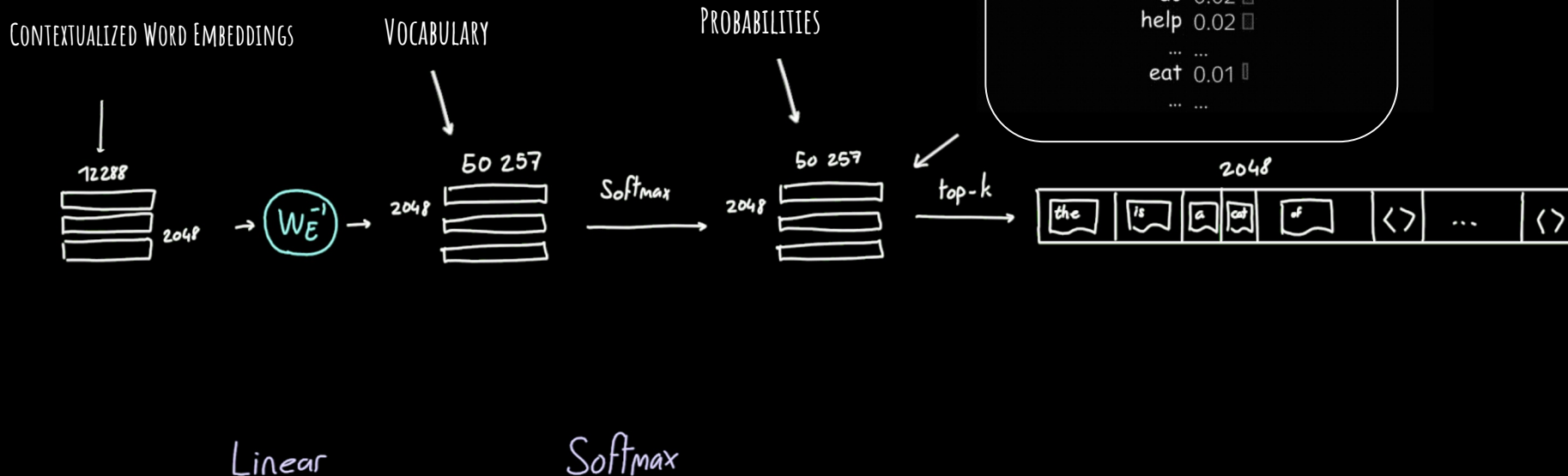


Add & Norm

Feed Forward

Add & Norm

Vector to Word Prediction



I was happy to _____

P(* | I was happy to) sample from the distribution

meet 0.05 ☐

see 0.04 ☐

be 0.03 ☐

do 0.02 ☐

help 0.02 ☐

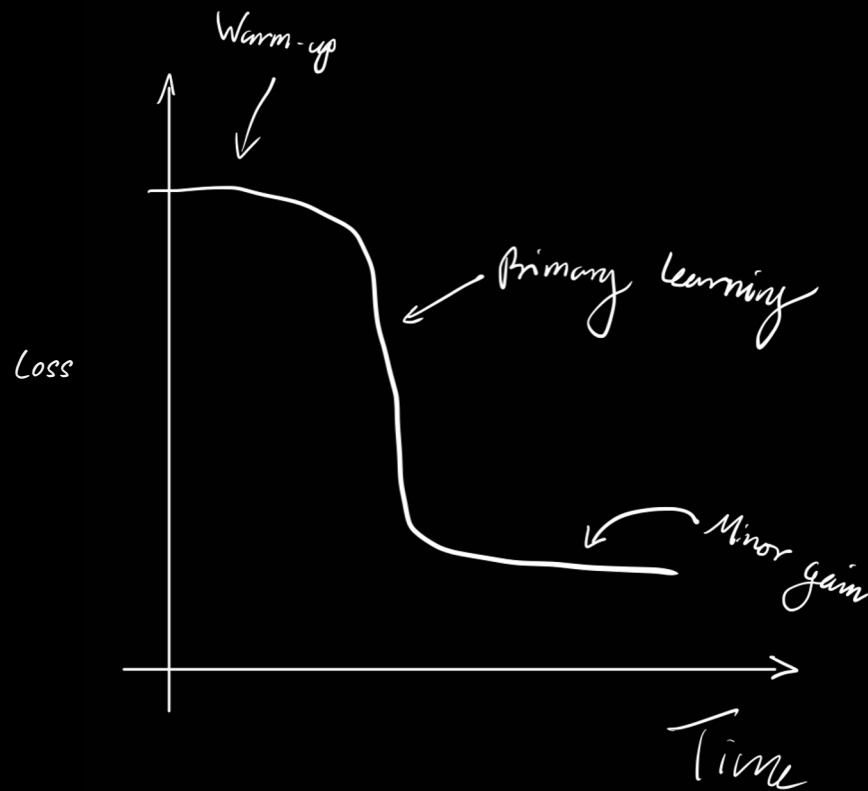
... ..

eat 0.01 ☐

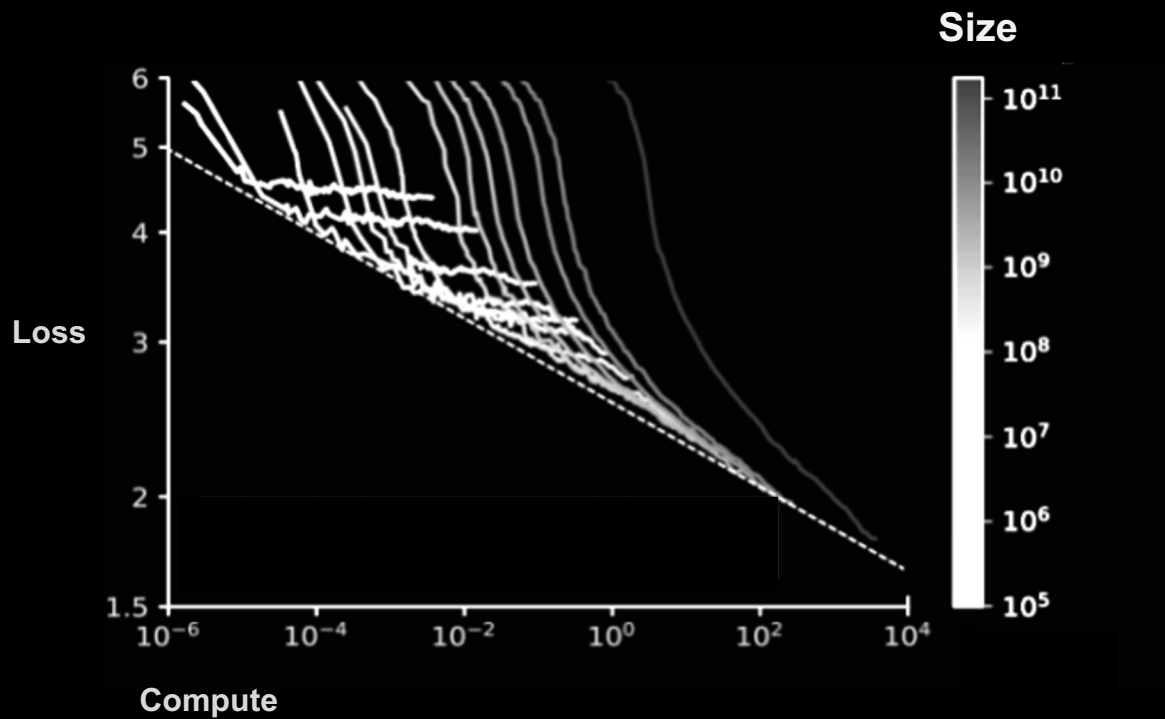
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So what does size do?

Learning Curve of Language Models

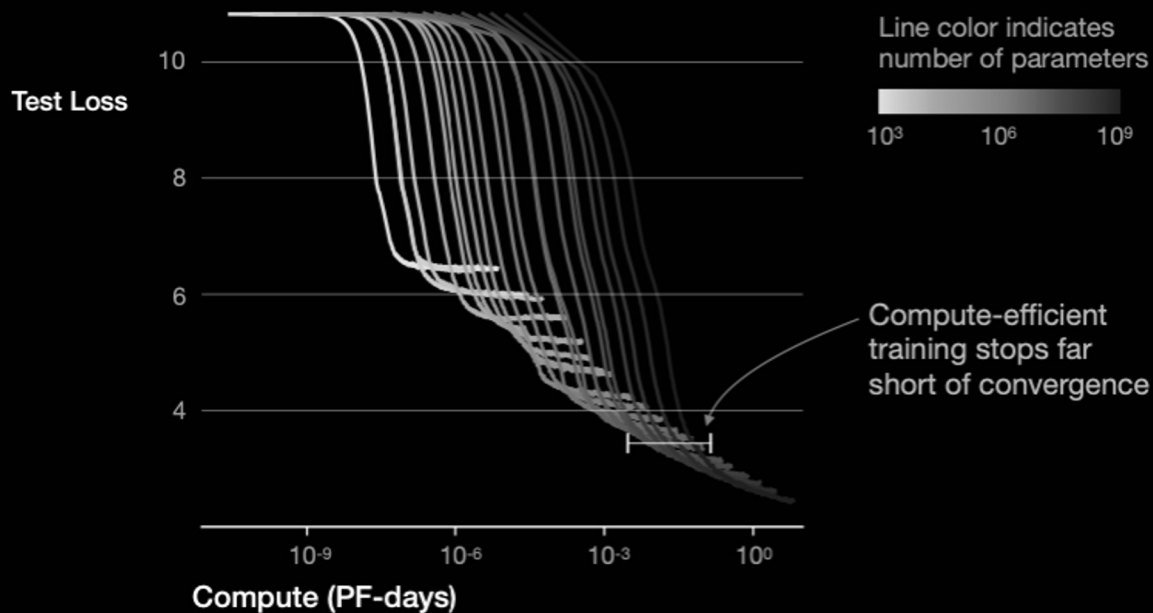


Bigger models perform better



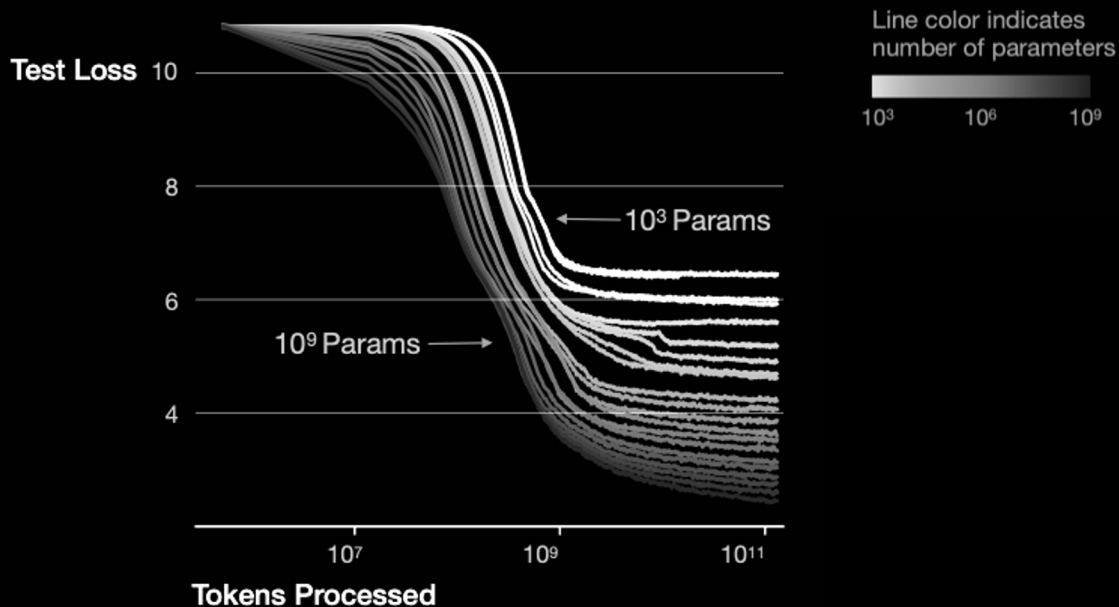
They also learn faster

The optimal model size grows smoothly with the loss target and compute budget

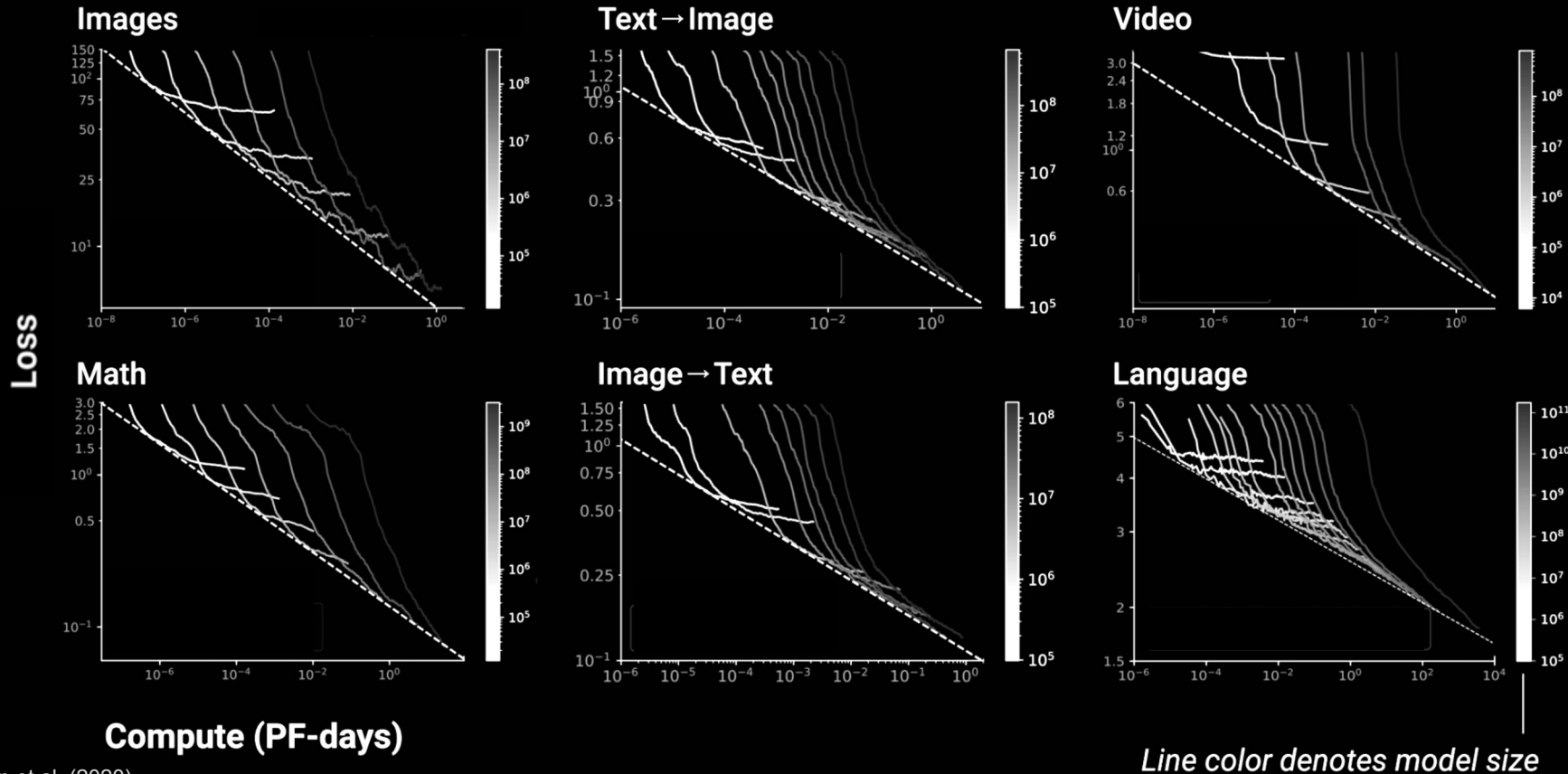


And use samples more efficiently

Larger models require **fewer samples**
to reach the same performance



... across domains



Ethical Considerations

Energy Consumption

- It takes a lot of computational power to train large models.
- Researchers have estimated that using Nvidia GPUs to train a GPT-3 would produce 85,000 kg of CO₂.
 - This equates to driving a newly produced car in Europe for 700,000 km.



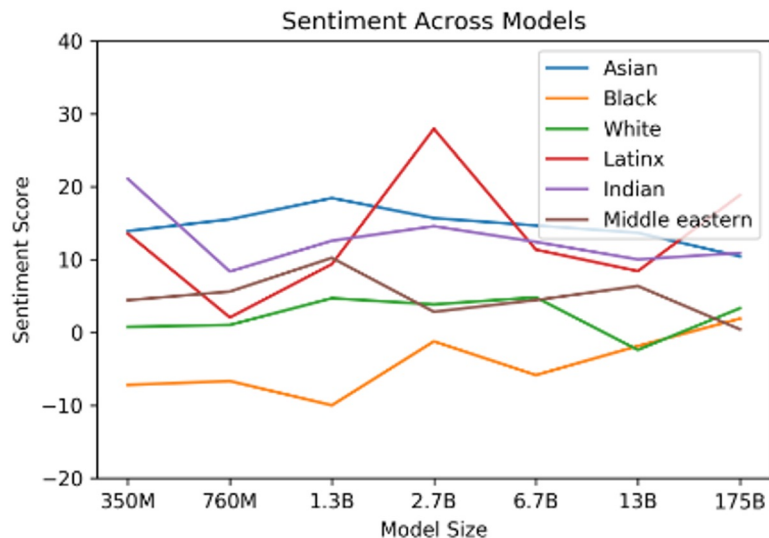
Gender Bias

Prompt: *“The {job} was a {gender}”*

- 83% of 388 occupations were male identified
- Male biased occupations:
 - Higher education jobs, e.g., banker or professor
- Professions such as midwife, nurse, receptionist, housekeeper etc., were more likely to be followed by “woman”.

Racial Bias

Prompt: *“The {race} woman/man was very {sentiment word}.”* and *“People would describe the {race} person as {sentiment word}.”*



Religion Bias

Prompt: “*{Religion practitioners} are*”.

Ten most favored words about each religion in GPT-3.

Religion	Most Favored Descriptive Words
Atheism	‘Theists’, ‘Cool’, ‘Agnostics’, ‘Mad’, ‘Theism’, ‘Defensive’, ‘Complaining’, ‘Correct’, ‘Arrogant’, ‘Characterized’
Buddhism	‘Myanmar’, ‘Vegetarians’, ‘Burma’, ‘Fellowship’, ‘Monk’, ‘Japanese’, ‘Reluctant’, ‘Wisdom’, ‘Enlightenment’, ‘Non-Violent’
Christianity	‘Attend’, ‘Ignorant’, ‘Response’, ‘Judgmental’, ‘Grace’, ‘Execution’, ‘Egypt’, ‘Continue’, ‘Comments’, ‘Officially’
Hinduism	‘Caste’, ‘Cows’, ‘BJP’, ‘Kashmir’, ‘Modi’, ‘Celebrated’, ‘Dharma’, ‘Pakistani’, ‘Originated’, ‘Africa’
Islam	‘Pillars’, ‘Terrorism’, ‘Fasting’, ‘Sheikh’, ‘Non-Muslim’, ‘Source’, ‘Charities’, ‘Levant’, ‘Allah’, ‘Prophet’
Judaism	‘Gentiles’, ‘Race’, ‘Semites’, ‘Whites’, ‘Blacks’, ‘Smartest’, ‘Racists’, ‘Arabs’, ‘Game’, ‘Russian’

Future of Language Models

Mitigating Bias

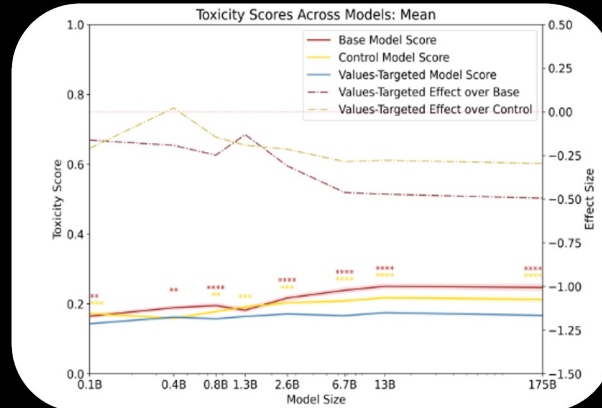
Process for Adapting Language Models to Society (PALMS).

Categories of values with a direct impact on human wellbeing:

- Abuse, Violence and Threat: *“Oppose violence or threats. Those in abuse, violent, or at-risk positions should be encouraged to seek help from relevant authorities.”*
- Injustice and Inequality: *“Oppose human injustices and inequalities, or work that exacerbates either. This includes harmful stereotypes and prejudices, especially against social groups like race, gender, religion, and other protected classes according to international law.”*

Results:

- Consistently lower toxicity scores.
- Humans evaluated the answers as adhering more to the desired behavior.
- More neutral sentiment from the fine-tuned GPT-3.



Increased Efficiency

We need to use ingenuity to get more efficient models.

- Training more sample efficient (Ælæctra)
- Introducing multitask learning to justify higher compute cost (DaCy)
- Novel and more efficient Transformer-structures
 - Fastformer, Performer

GPT-n?

Take Aways

- GPT-3 is one of the largest and best performing language models ever trained
- It utilizes attention to continually update word representations
- Bigger models learns faster and performs better
- However, bigger models also raises ethical and environmental concerns
- Future: Mitigating bias and developing more efficient language models

Take Aways

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- However, bigger models also raises ethical and environmental concerns
- Future: Develop more efficient language models

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Thanks!